

PROJECT REPORT

EXPERIMENT - 1

Project title

**Smart Elevator Automation and Human Detection System
Using PIR Sensor**

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Smart Elevator Automation and Human Detection System Using PIR Sensor

Project Overview

The Smart Elevator Automation and Human Detection System using PIR Sensor is a compact prototype designed to demonstrate automatic human detection and elevator floor selection using an ESP32 microcontroller. The system detects human movement using a PIR sensor and allows the user to select either Floor 1 or Floor 2 using push buttons.

The ESP32 acts as the central controller and processes the signals received from the PIR sensor and floor-selection buttons. Based on the detected human presence and selected floor, the ESP32 controls a servo motor to represent the movement of the elevator cabin.

Two LEDs are used to provide visual indications. LED 1 indicates that a human has been detected, while LED 2 indicates the elevator operating/status condition. A buzzer can provide an audible indication when the elevator reaches the selected floor or during a system event.

The proposed system provides a simple and low-cost demonstration of sensor-based elevator automation.

Main Objectives:

- Detect human presence using a PIR sensor.
- Use ESP32 as the main control unit.
- Provide floor selection using push buttons.
- Control elevator movement using a servo motor.
- Indicate human detection using LED 1.
- Indicate elevator status using LED 2.
- Provide audible indication using a buzzer.
- Demonstrate automatic elevator operation using a simple embedded system.

Problem Statement

Conventional basic elevator prototypes often depend mainly on manual floor-selection buttons and do not incorporate automatic human detection. This can result in elevator operation even when there is no person waiting to use the system.

Another challenge in a small elevator prototype is providing a simple and reliable mechanism for demonstrating cabin movement and floor selection. A conventional motor-based arrangement requires additional motor-driver circuitry, whereas a servo motor can provide controlled angular movement suitable for a compact demonstration model.

The proposed system addresses these issues by integrating a PIR sensor, ESP32 controller, push buttons, servo motor, LEDs and buzzer into a single automated elevator prototype.

When human movement is detected by the PIR sensor, the ESP32 activates the human-detection indication. The user can then select the required floor using the Floor 1 or Floor 2 push button. The ESP32 processes the selected input and commands the servo motor to move to the corresponding position. LEDs and a buzzer provide feedback about the system condition.

Problems Addressed:

➤ Lack of human detection

- Basic elevator systems may operate without detecting whether a person is present.
- The PIR sensor provides automatic human-motion detection.

➤ Manual floor selection

- The user can select the required floor using dedicated push buttons.
- The ESP32 processes the selected floor request.

➤ Lack of status indication

- LED 1 indicates human detection.
- LED 2 indicates elevator/system status.
- The buzzer provides an additional audible indication.

➤ **Complex motor arrangement**

- A servo motor is used instead of a DC motor and L298N driver.
- The servo can be directly controlled by the ESP32 signal pin.

➤ **Need for compact automation**

- The ESP32 integrates the control logic into a small microcontroller platform.
- The system can later be expanded with Wi-Fi and IoT features.

Proposed Solution

The proposed solution consists of an ESP32-based smart elevator control system.

The PIR sensor detects human movement and sends its output to GPIO 27 of the ESP32. Two push buttons are provided for floor selection:

- Push Button 1 → Floor 1
- Push Button 2 → Floor 2

The ESP32 processes the sensor and button inputs and controls a servo motor connected to GPIO 13.

The servo motor acts as the elevator movement actuator in the prototype. Different servo positions can represent different elevator floors.

Proposed Operation:

1. PIR sensor detects human movement.
2. PIR sends a HIGH signal to ESP32 GPIO 27.
3. ESP32 activates the Human Detected LED.
4. User presses Floor 1 or Floor 2 button.
5. ESP32 identifies the selected floor.
6. ESP32 commands the servo motor to move to the corresponding position.
7. Elevator status LED indicates the operation.
8. Buzzer provides an audible indication.
9. Servo reaches the predefined position representing the selected floor.
10. Elevator remains at the selected floor until another command is given.

System Architecture

The system consists of five main sections.

Input Section:

The input section consists of:

- PIR sensor
- Push Button 1 – Floor 1
- Push Button 2 – Floor 2

The PIR sensor detects human movement, while the push buttons allow the user to select the desired floor.

Processing Section:

The ESP32 acts as the central controller.

It receives:

- PIR sensor signal through GPIO 27
- Floor 1 button signal through GPIO 25
- Floor 2 button signal through GPIO 26

The ESP32 processes these inputs according to the programmed control logic.

Actuation Section:

A servo motor is connected to GPIO 13.

The servo motor receives a PWM control signal from the ESP32 and moves to predefined angular positions corresponding to the elevator floors.

For example:

- Floor 1 → Servo position 0°
- Floor 2 → Servo position 90°

The actual angles can be adjusted according to the mechanical arrangement of the prototype.

Indication Section:

The indication section contains:

- LED 1 – Human detected
- LED 2 – Elevator status
- Buzzer

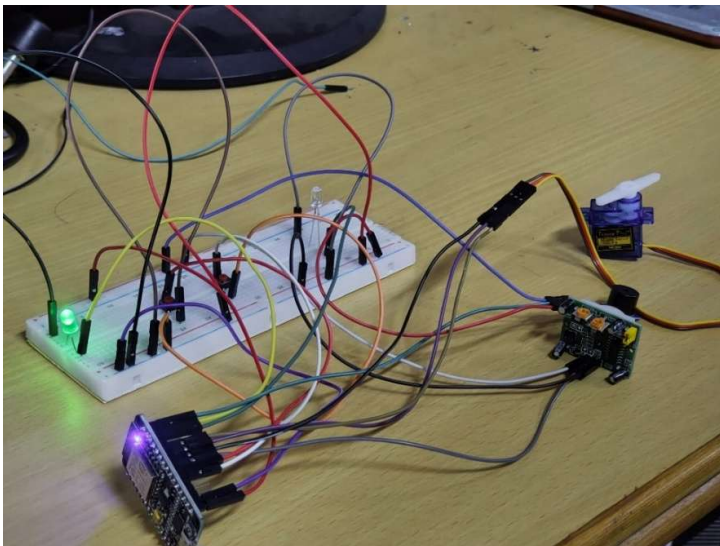
LED 1 provides visual confirmation when human movement is detected. LED 2 indicates elevator operation/status. The buzzer provides an audible indication when required.

Mechanical Section:

The mechanical section consists of:

- Miniature elevator cabin
- Guide/support structure
- Servo motor
- Connecting mechanism such as thread, arm or pulley arrangement
- Floor platforms

IMAGE:



Circuit table

CIRCUIT TABLE – SMART ELEVATOR SYSTEM (ESP8266)			
Component	Pin/Terminal	ESP8266 Connection	Function
 PIR Sensor	VCC	5V/VIN	Power supply
	GND	GND	Ground
	OUT	GPIO 14 (D5)	Human detection
 Push Button 1 – Floor 1	One terminal	GPIO 12 (D6)	Floor 1 selection
	Other terminal	GND	Ground
 Push Button 2 – Floor 2	One terminal	GPIO 13 (D7)	Floor 2 selection
	Other terminal	GND	Ground
 Servo Motor	Signal	GPIO 2 (D4)	Door control
	VCC	5V/VIN	Servo power
	GND	GND	Ground
 LED 1 – Door/ Elevator Operation	Anode (+) through 220 Ω	GPIO 5 (D1)	Indicates elevator operation
	Cathode (–)	GND	Ground
 LED 2 – Human Detection	Anode (+) through 220 Ω	GPIO 4 (D2)	Indicates person detected
	Cathode (–)	GND	Ground
 Buzzer	Positive (+)	GPIO 16 (D0)	Audible indication
	Negative (–)	GND	Ground

Important Pin Mapping:

GPIO 14 → D5 → PIR Sensor

GPIO 12 → D6 → Floor 1 Button

GPIO 13 → D7 → Floor 2 Button

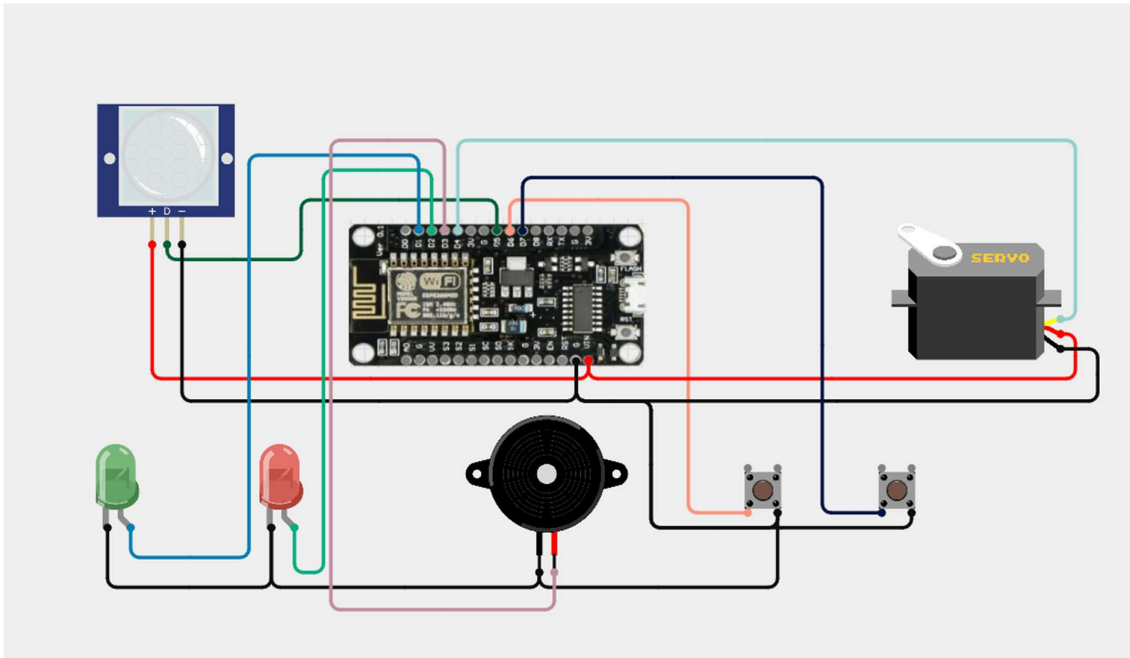
GPIO 5 → D1 → LED 1

GPIO 4 → D2 → LED 2

GPIO 16 → D0 → Buzzer

GPIO 2 → D4 → Servo Motor

Circuit Design:



Algorithm

Step-by-Step Algorithm

Step 1: Start the system.

Step 2: Initialize ESP32 GPIO pins.

Step 3: Initialize the servo motor.

Step 4: Set the elevator to the initial floor position.

Step 5: Read the PIR sensor.

Step 6: Check whether human movement is detected.

Step 7: If no human is detected, keep the elevator in standby mode.

Step 8: If human movement is detected, turn ON LED 1.

Step 9: Read Floor 1 and Floor 2 push buttons.

Step 10: Determine the selected floor.

Step 11: Move the servo to the corresponding predefined position.

Step 12: Turn ON LED 2 while the elevator is operating.

Step 13: When the servo reaches the required position, stop movement.

Step 14: Activate the buzzer to indicate that the destination has been reached.

Step 15: Maintain the elevator at the selected floor.

Step 16: Return to the monitoring state.

Coding

```
#include <Servo.h>
```

```
//———— Pin Definitions —————//
```

```
#define PIR_PIN 14 // D5
```

```
#define BUTTON1 12 // D6
```

```
#define BUTTON2 13 // D7
```

```
#define LED1 5 // D1
```

```
#define LED2 4 // D2
```

```
#define BUZZER 16 // D0
```

```
#define SERVO_PIN 2 // D4
```

```
Servo elevatorDoor;
```

```
//—————//
```

```
void setup()
```

```
{
```

```
  Serial.begin(115200);
```

```
pinMode(PIR_PIN, INPUT);
```

```
pinMode(BUTTON1, INPUT_PULLUP);
```

```
pinMode(BUTTON2, INPUT_PULLUP);
```

```
pinMode(LED1, OUTPUT);
```

```
pinMode(LED2, OUTPUT);
```

```
pinMode(BUZZER, OUTPUT);
```

```
digitalWrite(LED1, LOW);
```

```
digitalWrite(LED2, LOW);
```

```
// Active LOW buzzer
```

```
digitalWrite(BUZZER, HIGH); // OFF initially
```

```
elevatorDoor.attach(SERVO_PIN);
```

```
elevatorDoor.write(0);
```

```
Serial.println("Smart Elevator System Ready");
```

```
}
```

```
void loop()
```

```
{
```

```
// ————— PIR Indicator —————
```

```
if (digitalRead(PIR_PIN) == HIGH)
```

```
{
```

```
    digitalWrite(LED2, HIGH);
```

```
    Serial.println("Person Detected");
```

```
}
```

```
else
```

```
{
```

```
    digitalWrite(LED2, LOW);
```

```
}
```

```
//————— Button 1 —————
```

```
if (digitalRead(BUTTON1) == LOW)
```

```
{
```

```
    Serial.println("Floor 1 Selected");
```

```
    operateElevator();
```

```
    while (digitalRead(BUTTON1) == LOW); // Wait until released
```

```
    delay(200);
```

```
}
```

```
//————— Button 2 —————
```

```
if (digitalRead(BUTTON2) == LOW)
```

```
{
```

```
    Serial.println("Floor 2 Selected");
```

```
    operateElevator();
```

```
    while (digitalRead(BUTTON2) == LOW); // Wait until released
```

```
    delay(200);
```

```
}
```

```
}
```

```
//—————//
```

```
void operateElevator()
```

```
{  
  // LED1 ON  
  digitalWrite(LED1, HIGH);  
  
  // Active LOW Buzzer ON  
  digitalWrite(BUZZER, LOW);  
  
  Serial.println("Door Opening");  
  delay(3000);  
  
  // Buzzer OFF  
  digitalWrite(BUZZER, HIGH);  
  
  // Servo 0 -> 180  
  for (int pos = 0; pos <= 180; pos++)  
  {  
    elevatorDoor.write(pos);  
    delay(10);  
  }  
  // Keep door open  
  delay(5000);  
  
  Serial.println("Door Closing");  
  digitalWrite(BUZZER, LOW);  
  delay(2000);  
  digitalWrite(BUZZER, HIGH);  
  // Servo 180 -> 0  
  for (int pos = 180; pos >= 0; pos--)  
  {
```

```

    elevatorDoor.write(pos);

    delay(10);

}

// LED1 OFF

digitalWrite(LED1, LOW

}

```

Coding output:

The screenshot shows the Arduino IDE interface with the following code in the main editor:

```

25 pinMode(BUTTON1, INPUT_PULLUP);
26 pinMode(BUTTON2, INPUT_PULLUP);
27
28 pinMode(LED1, OUTPUT);
29 pinMode(LED2, OUTPUT);
30 pinMode(BUZZER, OUTPUT);
31
32 digitalWrite(LED1, LOW);
33 digitalWrite(LED2, LOW);
34
35 // Active LOW buzzer
36 digitalWrite(BUZZER, HIGH); // OFF initially
37
38 elevatorDoor.attach(SERVO_PIN);
39 elevatorDoor.write(0);
40
41 Serial.println("Smart Elevator System Ready");
42 }
43
44 void loop()
45 {

```

The Serial Monitor at the bottom shows the following output:

```

Floor 1 Selected
Door Opening
Door Closing
Floor 2 Selected
Door Opening
Door Closing

Person Detected
Person Detected
Person Detected
Person Detected

```

The status bar at the bottom indicates: Ln 50, Col 36 Generic ESP8266 Module on COM10.

System Working

The system operates using human detection, floor selection, servo-based door operation, LEDs and buzzer. The ESP8266 acts as the main controller.

Step 1 – System Initialization

When the system is switched ON:

- ESP8266 initializes all input and output pins.
- PIR sensor is configured as an input.
- Floor-selection buttons are configured using internal pull-up resistors.
- LEDs and buzzer are configured as outputs.
- Servo motor is attached to GPIO 2 (D4).
- Servo initially moves to 0°, representing the closed-door position.
- The serial monitor displays “Smart Elevator System Ready.”

Step 2 – Human Detection

The PIR sensor continuously monitors the surrounding area.

When a person is detected:

PIR Sensor



GPIO 14 (D5)



ESP8266



LED 2 ON



"Person Detected"

The ESP8266 turns LED 2 ON when the PIR output becomes HIGH.

When no person is detected:

- LED 2 is turned OFF.
- The system continues monitoring.

Step 3 – Floor 1 Selection

When the user presses Floor 1 Button:

- GPIO 12 (D6) becomes LOW.
- ESP8266 identifies that Floor 1 has been selected.
- The serial monitor displays:

“Floor 1 Selected”

The operate Elevator() function is then executed.

Step 4 – Floor 2 Selection

When the user presses Floor 2 Button:

- GPIO 13 (D7) becomes LOW.
- ESP8266 identifies that Floor 2 has been selected.
- The serial monitor displays:

“Floor 2 Selected”

The same operate Elevator() function is executed.

Elevator/Door Operation:

The operate Elevator() function controls the servo, LED and buzzer.

Stage 1 – Operation Indicator

When the elevator operation starts:

- LED 1 turns ON.
- Buzzer turns ON.
- Serial monitor displays:

“Door Opening”

The buzzer remains active for approximately 3 seconds.

Stage 2 – Door Opening

After the initial 3-second indication, the buzzer is turned OFF.

The servo moves gradually:

$0^{\circ} \rightarrow 180^{\circ}$

This movement represents the elevator door opening in the prototype.

The servo moves one degree at a time with a 10 ms delay between positions.

Stage 3 – Door Remains Open

After reaching 180° :

- The door remains open.
- The program waits for approximately 5 seconds.

This represents the time available for a passenger to enter or exit.

Stage 4 – Door Closing Indication

After the 5-second open period:

- Serial monitor displays “Door Closing.”
- Buzzer turns ON.
- It remains ON for approximately 2 seconds.
- The buzzer is then turned OFF.

Stage 5 – Door Closing

The servo moves in the reverse direction:

$180^{\circ} \rightarrow 0^{\circ}$

This represents the elevator door closing.

Once the servo reaches 0° :

- The door is considered closed.
- LED 1 is turned OFF.
- The elevator operation is completed.

Bill of Materials

S. No.	Component	Specification / Model	Quantity
1.	ESP8266 Node MCU	ESP8266 Wi-Fi Development Board	1
2.	PIR Motion Sensor	HC-SR501	1
3.	Servo Motor	SG90, 5 V	1
4.	Push Button – Floor 1	Momentary push button	1
5.	Push Button – Floor 2	Momentary push button	1
6.	LED 1	5 mm LED	1
7.	LED 2	5 mm LED	1
8.	Resistor	220 Ω	2
9.	Buzzer	3.3/5 V Active Buzzer	1
10.	Breadboard	400/830 point	1
11.	Jumper Wires	Male–Male / Male–Female	1
12.	Power Supply	5 V regulated supply	1
13.	USB Cable	Micro-USB	1
14	Elevator Prototype Structure	Cardboard/acrylic/3D printed	1

PROTOTYPE IMPLEMENTATION

The prototype of the Smart Elevator Automation and Human Detection System using ESP8266 can be implemented by connecting the PIR sensor, push buttons, LEDs, buzzer, and servo motor to the ESP8266 Node MCU. The system detects human movement, accepts a floor selection, provides an indication through LEDs/buzzer, and operates the servo motor as the elevator door.

Hardware Setup:

The components are assembled on a breadboard according to the following connections:

Component	Connection to ESP8266	Function
HC-SR501 PIR OUT	D5 (GPIO14)	Human detection
Push Button 1	D6 (GPIO12)	Select Floor 1
Push Button 2	D7 (GPIO13)	Select Floor 2
Servo Signal	D4 (GPIO2)	Elevator door
LED 1	D1 (GPIO5) through 220 Ω	Human detected
LED 2	D2 (GPIO4) through 220 Ω	Elevator/door status
Buzzer +	D0 (GPIO16)	Audio indication
PIR VCC	VIN/5 V	PIR power
Servo VCC	External regulated 5 V	Servo power
All GND	Common GND	Ground connection

Important: Connect the external 5 V servo supply ground to the ESP8266 GND.

Prototype Working:

The prototype operates in the following sequence:

Step 1 – Human Detection

When a person approaches the elevator, the PIR sensor detects motion and sends a HIGH signal to the ESP8266.

Step 2 – Human Indication

The ESP8266 switches LED 1 ON and activates the buzzer briefly to indicate that a person has been detected.

Step 3 – Floor Selection

The user presses one of the two push buttons:

- Button 1 → Floor 1
- Button 2 → Floor 2

The ESP8266 identifies the selected floor.

Step 4 – Elevator Door Operation

After a valid floor selection, the ESP8266 commands the servo motor to rotate to approximately 90°, representing the elevator door opening.

Step 5 – Door Closing

After a short delay, the servo returns to 0°, representing the elevator door closing. LED 2 can indicate the door/elevator operating status.

Expected Prototype Output:

Condition	LED 1	LED 2	Buzzer	Servo
No person	OFF	OFF	OFF	0°
Person detected	ON	OFF	Short beep	0°
Floor button pressed	ON	ON	—	90°
Door open	ON	ON	—	90°
Door closing	ON	ON	—	0°
Operation completed	ON/OFF	OFF	OFF	0°

Prototype Demonstration

For the physical demonstration, you can make a small cardboard/acrylic elevator model. Mount the servo at the elevator entrance and connect the servo arm to a lightweight door. Place the PIR sensor in front of the elevator so that movement is detected when someone approaches. The two push buttons can be labelled FLOOR 1 and FLOOR 2.

This gives you a simple working prototype demonstrating the main concept: human detection → floor selection → automatic elevator-door operation → status indication.

Results & Performance Analysis

The prototype is expected to detect human presence using the PIR sensor and automatically respond through the ESP8266. When a person is detected, the LED and buzzer provide an indication. The push buttons allow the user to select Floor 1 or Floor 2, and the servo motor operates the elevator door accordingly.

Expected Results

- PIR sensor successfully detects human movement.
- LED 1 indicates human presence.
- Buzzer provides an audible alert.
- Push buttons successfully select the required floor.
- Servo motor opens and closes the elevator door automatically.
- LED 2 indicates elevator/door operation.
- ESP8266 coordinates all sensors and outputs with quick response.

Performance Analysis

Parameter	Expected Performance
Human detection	Reliable within PIR sensing range
System response	Quick response after detection
Floor selection	Correct response to button presses
Servo operation	Smooth door opening/closing
Indication	Clear LED and buzzer alerts
Overall operation	Automatic and user-friendly